

# Clement LEE

*Address* School of Mathematics, Statistics and Physics  
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**Academic statistician with expertise in statistical programming and visualisation in R, and modelling of extremes and networks with computational Bayesian statistics.**

## WORK EXPERIENCE

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**Lecturer in Statistics** 2022-  
*School of Mathematics, Statistics and Physics, Newcastle University*  
Research and teaching. Past administrative role: seminar organiser.

**Lecturer in Statistics** 2019-2022  
*Department of Mathematics and Statistics, Lancaster University*  
Research and teaching. Award: ***Tower of Teaching*** for outstanding teaching in the department.

**Research Associate (Statistical Modeller)** 2016-2019  
*School of Mathematics and Statistics, Newcastle University*  
Undertaken statistical research using data generated by mobile and web technologies developed by the Open Lab, with a focus on social network and citation network analysis.

**Research Associate** 2015  
*Department of Mathematics and Statistics, Lancaster University*  
Derived the optimal scaling of one Markov chain Monte Carlo (MCMC) sampler, and implemented efficient MCMC algorithms for stochastic epidemic models based on such optimal scaling.

**Actuarial Intern** 2010  
*BOC Group Life Assurance Company Ltd., Hong Kong*  
Performed statistical analysis to assist pricing and valuation, and helped with checking the rates of new insurance products.

**Assistant in Actuarial Science** 2009  
*Department of Statistics and Actuarial Science, the University of Hong Kong*  
Developed an Excel software of insurance pricing for teaching and consulting.

## EDUCATION

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**PhD in Statistics** 2011-2015  
*Lancaster University*  
Thesis title: Statistical Models for Financial Time Series and the House Price Index.

**MSc in Statistics** 2010-2011  
*Lancaster University*  
Passed with Distinction. Awards: ***Chancellor's Medal***, and ***Royal Statistical Society prize*** for outstanding performance.

**BSc in Actuarial Science**

2006-2009

*The University of Hong Kong*

1st Class Honours. Awards: *HKU Worldwide Student Exchange Scholarship*, and *HKU Foundation Entrance Scholarship*.

## **PUBLICATIONS**

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### **Conditional Extremes with Graphical Models**

2027

A. Farrell, E. F. Eastoe and C. Lee

Computational Statistics & Data Analysis **225**, 108452

### **The Hierarchical Stochastic Block Model for Replicated Networks**

2026

M. Battiston and C. Lee

Statistics and Computing **36** (4), 180

### **A Bayesian Nonparametric Stochastic Block Model for Directed Acyclic Graphs**

2026

C. Lee and M. Battiston

Journal of Computational and Graphical Statistics **35** (2), 619–630

### **How to Utilise the Limited Supply of Vaccines for Mpox Control in Thailand among High-Risk GBMSM**

2026

S. Jirapanakorn, B. Hunsin, A. Endo, C. Lee, S. Jin, B. S. L. Dickens and T. Theparod

Infectious Disease Modelling **11** (3), 1009–1021

### **Evidencing Preferential Attachment in Dependency Network Evolution**

2025

C. Lee

arXiv preprint arXiv:2509.12135

### **Learning Growth Mechanisms of Tail Realistic Preferential Attachment Models from Network Degree Distributions**

2025

T. Boughen, C. Lee and V. Palacios Ramirez

arXiv preprint arXiv:2506.18726

### **Software Version Control Using Git: A Primer from UKRN**

2025

C. Lee et al.

OSF Preprint

### **Degree Distributions in Networks: Beyond the Power Law**

2024

C. Lee, E. Eastoe and A. Farrell

Statistica Neerlandica **78** (4), 702–718

### **Estimating the Asphaltene Critical Nanoaggregation Concentration Region Using Ultrasonic Measurements and Bayesian Inference**

2021

A. Svalova, D. Walshaw, C. Lee, V. Demyanov, N. G. Parker, M. J. Povey and G. D. Abbott

Scientific Reports **11**, 6698

### **A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets**

2020

C. Lee and D. J. Wilkinson

Journal of the American Statistical Association **115** (529), 1–15

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Performance and Sensitivities of Home Detection on Mobile Phone Data</b>                                                                                  | 2020 |
| M. Vanhoof, <u>C. Lee</u> and Z. Smoreda<br>Big Data Meets Survey Science: A Collection of Innovative Methods                                                |      |
| <b>A Review of Stochastic Block Models and Extensions for Graph Clustering</b>                                                                               | 2019 |
| <u>C. Lee</u> and D. J. Wilkinson<br>Applied Network Science 4 (1), 122                                                                                      |      |
| <b>Validity and Reliability of an Online Self-Report 24-h Dietary Recall Method (Intake24): A Doubly Labelled Water Study and Repeated-Measures Analysis</b> | 2019 |
| E. Foster, <u>C. Lee</u> , F. Imamura, S. E. Hollidge et al.<br>Journal of Nutritional Science, 8                                                            |      |
| <b>Weaving the Topics of CHI: Using Citation Network Analysis to Explore Emerging Trends</b>                                                                 | 2019 |
| <u>C. Lee</u> , A. Garbett, J. Wang, B. Hu and D. Jackson<br>Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems, 6          |      |
| <b>A Network Epidemic Model for Online Commissioning Data</b>                                                                                                | 2018 |
| <u>C. Lee</u> , A. Garbett and D. J. Wilkinson<br>Statistics and Computing 28 (4), 891-904                                                                   |      |
| <b>Optimal Scaling of the Independence Sampler: Theory and Practice</b>                                                                                      | 2018 |
| <u>C. Lee</u> and P. Neal<br>Bernoulli 24 (3), 1636-1652                                                                                                     |      |
| <b>A Social Network Analysis of Articles on Social Network Analysis</b>                                                                                      | 2018 |
| <u>C. Lee</u> and D. J. Wilkinson<br>arXiv preprint arXiv:1810.09781                                                                                         |      |
| <b>ThinkActive: Designing for Pseudonymous Activity Tracking in the Classroom</b>                                                                            | 2018 |
| A. Garbett, D. Chatting, G. Wilkinson, <u>C. Lee</u> and A. Kharrufa<br>Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, 7      |      |
| <b>LearningCircle.io: Lessons Learned from Organising Courses with and without a Dedicated Platform</b>                                                      | 2018 |
| H. Celina, <u>C. Lee</u> , P. Olivier and A. Kharrufa<br>EdMedia+ Innovate Learning, 76-81                                                                   |      |

## **FUNDING SUCCESSES**

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| <b>Knowledge Exchange Placement</b>                                                                                                       | 2026 |
| <i>Funder: DICE Network+</i><br>Project: Predicting item desirability with machine learning to boost digital platform exchanges           |      |
| <b>Knowledge Exchange Catalyst</b>                                                                                                        | 2025 |
| <i>Funder: International Centre for Mathematical Sciences</i><br>Project: Turbo-charging the gift economy through geospatial optimisation |      |

## PHD SUPERVISION

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**Clive Drowley**

2024-

*Geospatial Centre for Doctoral Training, Newcastle University*

First supervisor; Topic on geospatial analysis of network flows

**Thomas Boughen**

2023-

*School of Mathematics, Statistics and Physics, Newcastle University*

First supervisor; Topic on intersections of extreme value theory, networks & Bayesian nonparametrics

**Aiden Farrell**

2021-2025

*Department of Mathematics and Statistics, Lancaster University*

Second supervisor; Topic on the extremal behaviour of the nodes when a network grows

**Maarten Vanhoof**

2017-2018

*School of Computing, Newcastle University*

Second supervisor; Thesis title: Geographical Veracity of Indicators from Mobile Phone Data

## TEACHING

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**Undergraduate and Postgraduate Teaching**

2023-

*School of Mathematics, Statistics and Physics, Newcastle University*

Taught modules for BSc and MSc students in Mathematics, Statistics & Data Science, including *Statistical Foundations of Data Science*, *Statistical Learning Methodology*, and *Modern Bayesian Inference*. Developed and delivered new module on *Data Visualisation*.

**Postgraduate Teaching**

2019-22

*Department of Mathematics and Statistics, Lancaster University*

Taught modules for MSc students in Statistics and in Data Science, including *R programming*, *Statistical Inference*, *Generalised Linear Models*, and *Extreme Value Theory*.

**Graduate Teaching Assistant**

2011-2015

*Department of Mathematics and Statistics, Lancaster University*

Marked assignments, held workshop sessions, and provided help and feedback to students. Modules included *Bayesian Inference*, *Likelihood Inference*, *Time Series Analysis*, and *Stochastic Processes*.

## PROFESSIONAL ACTIVITIES

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**Statisticians for Society Scheme**

2024-

*Royal Statistical Society*

Member of review panel for *pro bono* consultancy projects carried out by fellow RSS statisticians for charity organisations.

**Local Lead on Open Research**

2024-

*UK Reproducibility Network (UKRN) & Newcastle University*

Led grassroots activities on open research supported by UKRN, at Newcastle University, including the university's inaugural Open Research Conference, training workshops on computational reproducibility, and a journal club on open research.

## INVITED TALKS

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|                                                                                                                                                                         |            |
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| <b>Community-Powered Circular Economy: The Role of Data</b><br><i>Exeter Centre for Circular Economy</i>                                                                | 2026-05-14 |
| <b>Evidencing Preferential Attachment in Dependency Network Evolution</b><br><i>Statistics Seminar, Department of Mathematics, University of York</i>                   | 2026-03-19 |
| <b>Breaking Barriers: Enhancing ECR Engagement in Open Science</b><br><i>ReproducibiliTea Journal Club, University of York</i>                                          | 2025-03-28 |
| <b>Lessons from My (S)crappy Reproducibility Journey</b><br><i>Royal Statistical Society 2024 Annual Conference</i>                                                     | 2024-09-04 |
| <b>Citation Network Analysis of a Computer Science Conference</b><br><i>The Bartlett Centre for Advanced Spatial Analysis, UCL</i>                                      | 2020-02-28 |
| <b>A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets</b><br><i>Department of Statistics and Actuarial Science, University of Hong Kong</i> | 2019-01-02 |

## SKILLS

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|----------------------|----------------------------------------------------------------|
| <i>Programming</i>   | R, C++ and Python (elementary)                                 |
| <i>Text editing</i>  | EMACS, L <sup>A</sup> T <sub>E</sub> X, and (R)Markdown        |
| <i>Miscellaneous</i> | Git and Bash                                                   |
| <i>Languages</i>     | English, Cantonese Chinese (native), Mandarin Chinese (fluent) |